

Atle Guttormsen, PhD School of Economics and Business Norwegian University of Life Sciences (NMBU)
UniversitetstUNET 3, 1433 Ås, Norway atle.guttormsen@nmbu.no

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Editors-in-Chief *Psychology of Sport and Exercise*

Dear Editors,

I am pleased to submit the enclosed manuscript, **“Pulling back before dropout: Behavioral disengagement precedes youth-sport exit by years in a 14-year register study,”** for consideration as an original research article in *Psychology of Sport and Exercise*.

The paper addresses a question central to sport psychology and youth sport policy: *why do adolescents leave organized competitive sport, and can the process be observed before it concludes?* The qualitative and survey-based literature has long argued that dropout is a gradual deliberative process rather than a single decision (Eliasson & Johansson, 2021; Espedalen & Seippel, 2022). Yet the *longitudinal behavioral signature* of this process — what gradual disengagement actually looks like in the year-by-year activity of athletes who will eventually leave — has been hard to observe directly. Surveys lose dropped-out athletes through selection bias; interviews recover the process only retrospectively.

This manuscript uses the Norwegian Athletics Federation’s complete public competition register to bridge that gap. I followed all 2,123 athletes who participated in six consecutive editions of a regional youth track-and-field meet (2011–2016) for up to 14 years, recording each athlete’s annual competition volume, event diversification, championship participation, and Tyrving age-norm performance. The 14-year follow-up and complete-population design eliminate the recall and selection biases that limit prior dropout research.

Principal findings. Baseline-window competition volume (ages 13–14) provided the largest incremental contribution to prospective discrimination of senior retention in a baseline-only logistic regression (OR = 2.40 per SD, 95% CI [2.07, 2.78]; cross-validated AUC = 0.751). Effects were unchanged with structural controls (region, birth quarter, club size all non-significant). The behavioral signal decomposes into two substantial and independent components: baseline behavioral heterogeneity (volume at age 14, OR = 2.79) and a within-athlete pull-back across the age-14-to-15 transition (change OR = 2.44; pseudo- R^2 doubled from 0.11 to 0.23). Findings replicated across two independent birth-year cohorts (1998–2000 and 2001–2002).

Fit with PSE. The manuscript fits *Psychology of Sport and Exercise* on three grounds. First, the theoretical scaffolding spans the Sport Commitment Model (Scanlan et al., 1993, 2016) — Scanlan’s SCQ-2 was published in PSE itself — supplemented by Self-determination theory (Ryan & Deci, 2000; Sarrazin et al., 2002; Vallerand, 2007) and the withdrawal-as-process tradition (Eliasson & Johansson, 2021; Espedalen & Seippel, 2022). Second, the design supplies the kind of *longitudinal behavioral* evidence missing from a literature dominated by cross-sectional surveys and short-prospective designs. Third, the practical implications — federations and clubs can use register-based behavioral surveillance (PPV $\approx$ 0.94 at a moderate-sensitivity threshold) to identify at-risk athletes years before formal exit — are directly actionable for the applied audiences PSE serves.

Methodological rigor. Every effort has been made to anticipate the methodological concerns that arise with observational survival analysis at this scale. The proportional-hazards assumption was tested via Schoenfeld residuals and addressed where violated by period-specific Cox modelling (Table 5). A landmark analysis at age 16 (Supplementary Table S8) directly tests for measurement overlap between predictor and outcome — the dominant effect (HR = 0.60) survives. Sensitivity to unmeasured confounding is quantified via E-value (E = 5.16). Estimates are robust to outcome definition (Table S9), clustering by club (Table S2), missing-data treatment (Table S5), exclusion of zero-volume athletes (Table S11), cross-cohort replication (Table 7), and sex-stratified subgroup analyses (Table S7). A specialization-vs-performance confound check is reported in Table S18. The behavioral signal is described throughout as a *marker* of progressive disengagement rather than a direct measurement of motivation — an empirical contribution to a theoretical conversation, not a theoretical claim in itself.

A note on a-priori power. The journal’s guidance asks for a-priori power estimates “drawing on effect sizes from previous meta-analyses and/or specific papers with designs close to the one authors use.” Because the present design uses the complete population meeting inclusion criteria — with no sampling step at which the analyst chooses an N — a classical a-priori power calculation is not directly applicable. Methods Section 2.5.5 reframes this requirement as a *detection-capacity* analysis: the minimum detectable hazard ratio given the observed event count is $HR \approx 1.07$ per SD (Hsieh & Lavori, 2000), well below typical effect-size magnitudes in prior prospective youth-sport dropout research (Cohen’s $d \approx 0.2$ – 0.5 ; ORs ≈ 1.4 – 2.5 ; Calvo et al., 2010; Espedalen & Seippel, 2022; Sarrazin et al., 2002). I hope this reframing satisfies the journal’s underlying intent that the design be capable of detecting theoretically meaningful effects.

Compliance. The main manuscript runs approximately 7,700 words including the abstract, references, and the AI declaration, with seven main tables and three figures. This is comfortably within the journal’s 8,000-word target. Eighteen supplementary tables (S1–S18) and six supplementary figures (S0–S5) are housed in supplementary material. Analysis follows STROBE reporting guidance for observational studies (von Elm et al., 2007) and SAGER guidance for sex/gender reporting (Heidari et al., 2016). The underlying competition records are publicly accessible via the Norwegian Athletics Federation’s online register; the derived analysis dataset itself contains personal data (athlete dates of birth) under the EU GDPR and is therefore not publicly redistributable, although the full analysis code will be deposited upon acceptance. No individual is identifiable in the analysis or the manuscript. The submission is prepared for double-anonymized peer review, with the title page submitted separately. Disclosure of generative-AI use in manuscript preparation appears in a dedicated section before the references list, per the journal’s policy. I confirm that the manuscript is not under consideration elsewhere and that I have no competing interests.

I look forward to receiving the editors’ and reviewers’ feedback. Thank you for considering this work.

Sincerely,

Atle Guttormsen, PhD School of Economics and Business Norwegian University of Life Sciences (NMBU)
atle.guttormsen@nmbu.no